

K-FIELD LATERAL FORCE ANALYSIS

Exact Continuous-Rim Integration for a 1 kg, 0.10 m Radius Uniform Flywheel

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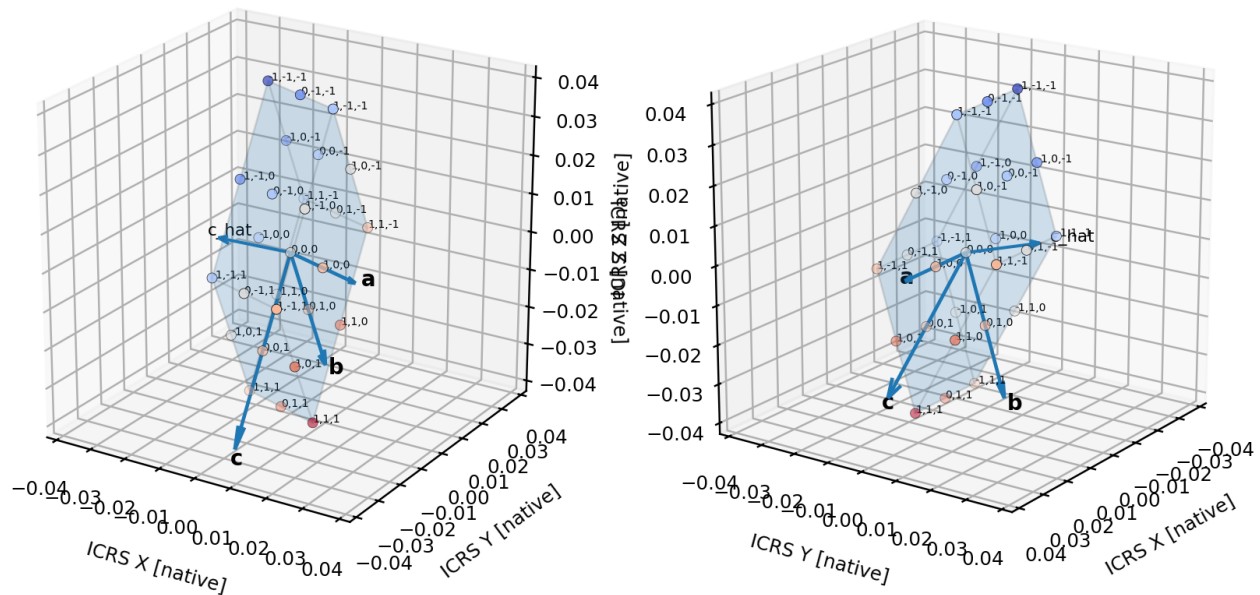
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Origin-locked ICRS Cell3 base cell: complete node and boundary geometry

View 1: all 27 half-step nodes

View 2: all 27 half-step nodes



Title figure. The complete origin-locked ICRS Cell3 base cell, including all 27 centered half-step nodes, the six boundary faces, the primitive basis vectors, and the Earth-CMB direction. Every coordinate is generated directly from the supplied machine-readable geometry.

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Abstract

This reference develops the exact lateral-force response of a one-kilogram, uniform, zero-thickness flywheel rim of radius 0.10 m in the locked ICRS K-Field geometry. The flywheel translates with the Earth-CMB velocity state and rotates about one fixed axis. The elemental force is integrated over the complete continuous mass distribution rather than approximated by isolated point masses. Antipodal rim elements reverse both their radial direction and their translation-coupling sign, so their lateral contributions add. The radius-dependent rotational self-term cancels exactly. The surviving resultant is $F_{\text{ring}} = (\kappa_F M \omega / 2) \mathbf{n} \times \mathbf{G}_{\text{phase}} \mathbf{V}_E$. When the axis $\mathbf{n}$ equals the Earth-CMB direction $\mathbf{c}_{\text{hat}}$, the result is a steady force perpendicular to $\mathbf{c}_{\text{hat}}$. For $M=1$ kg and $\omega=1$ rad/s, the Earth-calibrated force is 7.5053656 microN with calibration interval 7.4289192 to 7.5818120 microN. The force direction is RA 84.0821143 deg and Dec 41.2899223 deg. A further geometric optimization shows that the maximum steady force that remains purely lateral is obtained by choosing the spin axis perpendicular to $\mathbf{G}_{\text{phase}} \mathbf{V}_E$ within the plane spanned by $\mathbf{c}_{\text{hat}}$ and $\mathbf{G}_{\text{phase}} \mathbf{V}_E$. This raises the steady force to 14.3070205 microN at the same mass and angular rate. The report supplies the complete Cell3 node and plane geometry, the exact integral, calibrated numerical results, scaling laws, mechanical loads, reconstruction tests, and machine-readable ledgers.

Field	Authoritative or derived value
Coordinate frame	ICRS/J2000 Cartesian; origin at Earth center
Flywheel	Uniform zero-thickness rim; $M=1.0$ kg; $R=0.10$ m; $D=0.20$ m
Reference spin	$\omega=1.0$ rad/s = 9.549296586 rpm
Earth-CMB vector	$[-359003.199066, 76679.886034, -44701.060101]$ m/s
Earth coupling	$\kappa_F=3.074050899702\text{e-}11$; interval $3.042739944755\text{e-}11$ to $3.105361854649\text{e-}11$
Aligned-axis exact result	$F=(\kappa_F M \omega / 2) \mathbf{c}_{\text{hat}} \times \mathbf{G}_{\text{phase}} \mathbf{V}_E$
Aligned-axis force	7.505365563 microN, steady and purely lateral
Maximum pure-lateral force	14.307020483 microN at 58.359138729 deg axis tilt from $\mathbf{c}_{\text{hat}}$
Torque	Zero at every rim element because force is radial
Machine-readable companion	K-Field_laerl_force_analysis_1784896442.json

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1. Purpose, Scope, and Result Hierarchy

The calculation addresses a single-axis continuous flywheel rather than the earlier two-axis point-mass assembly. The requested body is a circular line mass: one kilogram is distributed uniformly around a circle of radius 0.10 m, and the rim has zero radial and axial thickness in the mathematical model. The center is fixed at the ICRS origin used by the source geometry. The body shares the Earth-CMB translation state and carries an added rotational velocity.

The result hierarchy is deliberately explicit. First, the complete element law is written with both translation-coupled and rotational self terms. Second, antipodal symmetry removes the self term and shows why opposite rim elements add rather than cancel. Third, the remaining force is integrated over the full circumference. Fourth, the Earth-derived coupling constant converts the unit-coupling coefficient into engineering force. Fifth, the spin-axis orientation is examined to distinguish the convenient CMB-aligned configuration from the maximum pure-lateral configuration.

Quantity	Symbol	Value
Total rim mass	M	1.000000 kg
Radius	R	0.100000 m
Diameter	D	0.200000 m
Circumference	2 pi R	0.628318530718 m
Linear mass density	$M/(2 \pi R)$	1.591549430919 kg/m
Moment of inertia	$I=M R^2$	0.010000000000 kg m ²
Reference angular rate	omega	1.000000 rad/s
Reference spin rate	RPM	9.549296585514 rpm
Reference rim speed	R omega	0.100000000000 m/s
Reference rotational energy	$0.5 I \omega^2$	0.005000000000 J
Reference angular momentum	I omega	0.010000000000 kg m ² /s
Ideal hoop tension	$M R \omega^2/(2 \pi)$	0.015915494309 N

2. Coordinate Lock and Constitutive Inputs

All vectors are expressed in the fixed ICRS Cartesian frame. The orthonormal triad $\mathbf{e}_0$, $\mathbf{A}_0$, and $\mathbf{c}_{\text{hat}}$ is right-handed. The vector $\mathbf{c}_{\text{hat}}$ is the Earth-CMB translation direction, while $\mathbf{e}_0$ and $\mathbf{A}_0$ span its perpendicular plane. The identity $\mathbf{e}_0 \text{ cross } \mathbf{A}_0 = \mathbf{c}_{\text{hat}}$ fixes the sign of every subsequent cross product.

Vector	ICRS components	RA [deg]	Dec [deg]
$\mathbf{c}_{\text{hat}}$	[-0.970770855836, 0.207348009111, -0.120874929482]	167.943299880	-6.942599922
$\mathbf{e}_0$	[-0.200864884110, -0.426231479950, 0.882031758970]	244.767479396	61.888432983
$\mathbf{A}_0$	[0.131366829113, 0.880530254232, 0.455422032396]	81.514594988	27.092091822
$\mathbf{G}_{\text{phase}} \mathbf{V}_E$	[-880221.981635, -143911.436562, 266335.026949]	189.285391176	16.626277960

$\mathbf{e}_0 \cdot \mathbf{A}_0 = 0 \quad \mathbf{e}_0 \cdot \mathbf{c}_{\text{hat}} = 0 \quad \mathbf{A}_0 \cdot \mathbf{c}_{\text{hat}} = 0 \quad \mathbf{e}_0 \text{ cross } \mathbf{A}_0 = \mathbf{c}_{\text{hat}}$

The phase metric $\mathbf{G}_{\text{phase}}$ transforms the Earth-CMB velocity into the anisotropic kernel vector $\mathbf{g} = \mathbf{G}_{\text{phase}} \mathbf{V}_E$. The flywheel force depends on this transformed vector, not on $\mathbf{V}_E$ alone.

Input	Value
$\mathbf{G}_{\text{phase}}$ row 1	[2.9764106378, 1.2431507795, -2.0803600354]
$\mathbf{G}_{\text{phase}}$ row 2	[1.2431507795, 2.5922286635, -2.3178839937]
$\mathbf{G}_{\text{phase}}$ row 3	[-2.0803600354, -2.3178839937, 6.7735709133]
$\mathbf{V}_E$ [m/s]	[-359003.199065813038, 76679.886034001989, -44701.060100687995]
$ \mathbf{V}_E $	369812.502000406 m/s
$\mathbf{g} = \mathbf{G}_{\text{phase}} \mathbf{V}_E$	[-880221.981635091, -143911.436561599, 266335.026948619] m/s

Input	Value
g	930825.217270435 m/s
angle(c_hat,g)	31.640861271 deg

The dimensionless Earth coupling is obtained from the two Earth-kinematic channels stored in the calibration source. The recommended value is the midpoint of the vector-projection and apsidal-rate values.

Calibration channel	kappa_F
Earth eccentricity-vector projection	3.04273994475465e-11
Earth apsidal rate	3.10536185464942e-11
Recommended Earth coupling	3.07405089970203e-11
Resolved interval	3.04273994475465e-11 to 3.10536185464942e-11

3. Authoritative Cell3 Base Geometry

The force calculation uses the same origin-locked Cell3 orientation as the Earth coupling derivation. The primitive cell is generated from the three non-orthogonal basis vectors a, b, and c. Twenty-seven half-step nodes occupy the coefficient set $\{-0.5, 0, +0.5\}$ in each primitive direction. Six boundary planes close the cell, two for each primitive face family.

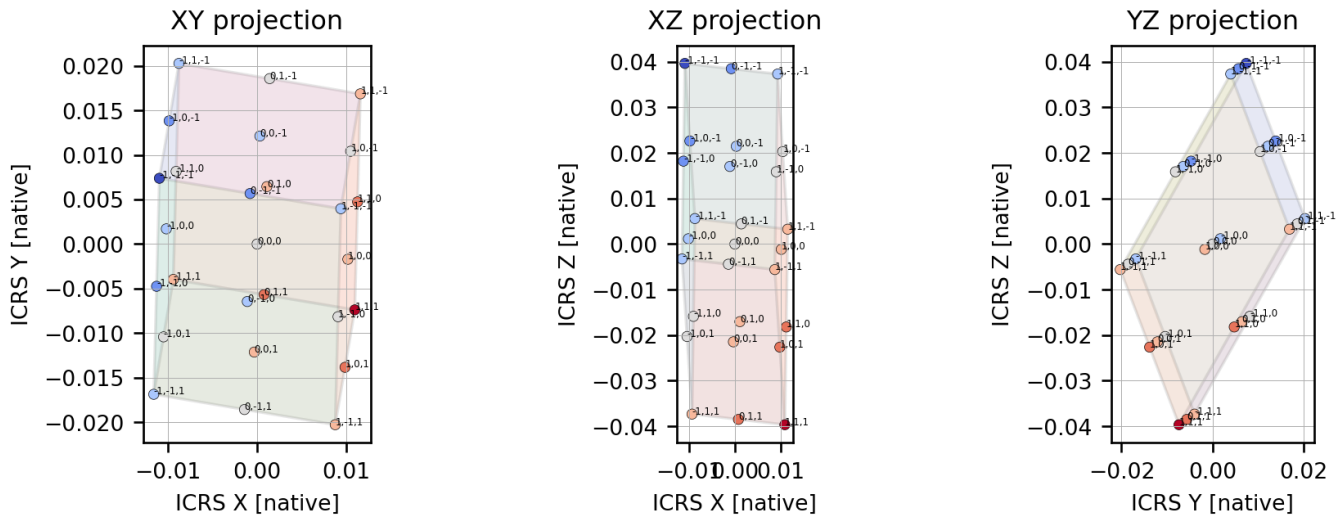


Figure 3.1. Orthographic projections of the dimensionally locked Cell3. Every node label gives the integer half-step index (i,j,k), where the physical coefficient is one half of that integer. The translucent polygons are the six boundary faces.

Primitive	ICRS basis vector [native]	Length [native]
a	[0.020381438517, -0.003416414496, -0.002332965917]	0.020797058782
b	[0.002198745968, 0.012902170259, -0.034066501536]	0.036494205131
c	[-0.000630379485, -0.024257722017, -0.042924354571]	0.049308565900
Cell volume	det[a b c]	2.842066937809649e-05 native^3

The face areas and normals identify the three plane families that bound the cell. The plane equations use $\mathbf{n} \cdot \mathbf{x} + d = 0$ in native ICRS coordinates.

Face	Area [native^2]	Included angle [deg]	Unit normal [ICRS]
A_ab	7.547217896726240e-04	83.933483012	[0.194092201, 0.913177169, 0.358379235]
A_ac	1.011255757688065e-03	80.448130018	[0.089052594, 0.866576769, -0.491033948]
A_bc	1.385784431733287e-03	50.363231814	[-0.995965190, 0.083602163, -0.032619291]

4. Flywheel Geometry and Continuous Mass Measure

The flywheel is parameterized in its own plane. For the CMB-aligned reference configuration, the spin axis is $\mathbf{n}=\mathbf{c_hat}$, so the rim plane is spanned by $\mathbf{e0}$ and $\mathbf{A0}$. The radial unit vector $\mathbf{u}(\theta)$ points from the center to the rim, and the tangent $\mathbf{t}(\theta)$ points in the direction of increasing θ .

$$\begin{aligned} \mathbf{u}(\theta) &= \cos(\theta)\mathbf{e0} + \sin(\theta)\mathbf{A0} & \mathbf{t}(\theta) &= -\sin(\theta)\mathbf{e0} + \cos(\theta)\mathbf{A0} = \mathbf{c_hat} \times \mathbf{u}(\theta) \\ r(\theta) &= R & dm &= (M/2\pi)d\theta \end{aligned}$$

Requested flywheel: 1 kg uniform zero-thickness rim, $R = 0.10$ m

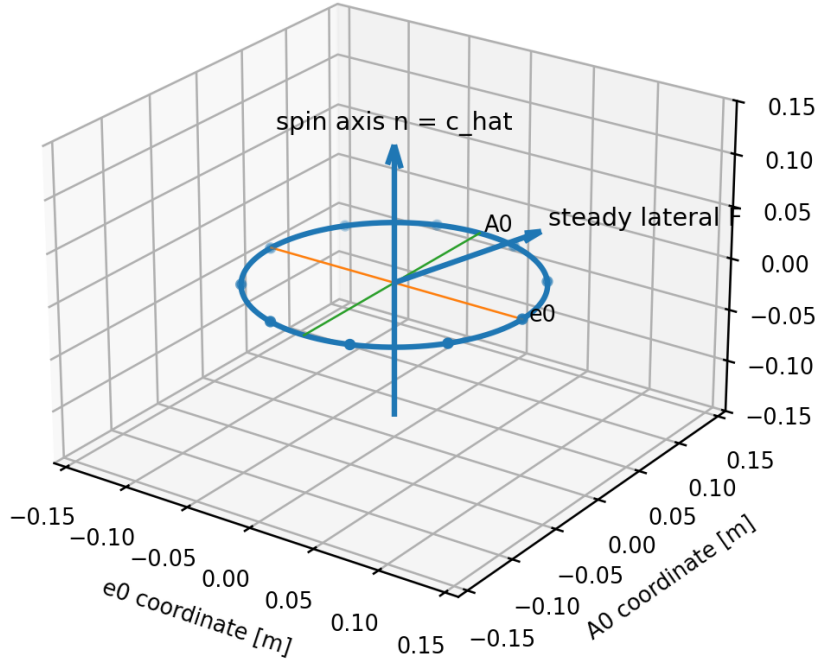


Figure 4.1. Dimensionally correct 0.10 m radius rim in the local $\mathbf{e0}$ - $\mathbf{A0}$ - $\mathbf{c_hat}$ frame. The steady force arrow is drawn in the measured ICRS force direction but scaled geometrically for visibility.

Uniform occupancy is essential. The mass per unit angle is $M/(2\pi)$, independent of radius. The linear density is $M/(2\pi R)$, so reducing radius raises the material density per unit circumference but does not change the total angular mass measure.

5. Elemental K-Field Force Law

Each rim element has the common Earth-CMB velocity $\mathbf{V_E}$ plus its local tangential velocity $R\omega\mathbf{t}(\theta)$. The anisotropic phase metric produces two element-level kernels.

$$\begin{aligned} \mathbf{v}(\theta) &= \mathbf{V_E} + R\omega\mathbf{t}(\theta) & \mathbf{B}(\theta) &= \mathbf{t}(\theta)^T \mathbf{G_phase} \mathbf{V_E} \\ \mathbf{C}(\theta) &= \mathbf{t}(\theta)^T \mathbf{G_phase} \mathbf{t}(\theta) \end{aligned}$$

$\mathbf{B}(\theta)$ is the translation-rotation cross kernel. It has velocity units. $\mathbf{C}(\theta)$ is the dimensionless rotational self kernel. The radial force on an element is

$$d\mathbf{F}(\theta) = -\kappa_F dm [\omega \mathbf{B}(\theta) + R\omega^2 \mathbf{C}(\theta)]\mathbf{u}(\theta)$$

Dimensional closure is direct: $\omega \mathbf{B}$ has units $1/s \times m/s = m/s^2$; $R\omega^2 \mathbf{C}$ also has m/s^2 . Multiplication by dm gives newtons. The force is radial because it is proportional to $\mathbf{u}(\theta)$. Consequently, the torque about the flywheel center is zero at every element, not merely after averaging.

$$d\boldsymbol{\tau} = \mathbf{r}(\theta) \times d\mathbf{F}(\theta) = R\mathbf{u}(\theta) \times [\text{scalar times } \mathbf{u}(\theta)] = \mathbf{0}$$

6. Antipodal Addition and Exact Self-Term Cancellation

The point opposite theta is theta+pi. Its radial and tangential vectors both reverse:

$$u(\theta+\pi) = -u(\theta) \quad t(\theta+\pi) = -t(\theta) \quad B(\theta+\pi) = -B(\theta) \quad C(\theta+\pi) = C(\theta)$$

Substitution into the full element law shows the sign structure. The radius-dependent self term acts oppositely on the two sides and cancels. The translation-coupled term changes sign at the same time that the radial direction changes sign, so the physical force direction is preserved.

$$dF(\theta) + dF(\theta+\pi) = -2 \kappa_F dm \omega B(\theta) u(\theta)$$

In common language, opposite rim elements do not cancel because two reversals occur together. The element on the far side moves in the opposite tangential direction, which reverses its K-field response. It is also located in the opposite radial direction. Reversing both the response sign and the radial direction makes its force point the same way as the near-side force.

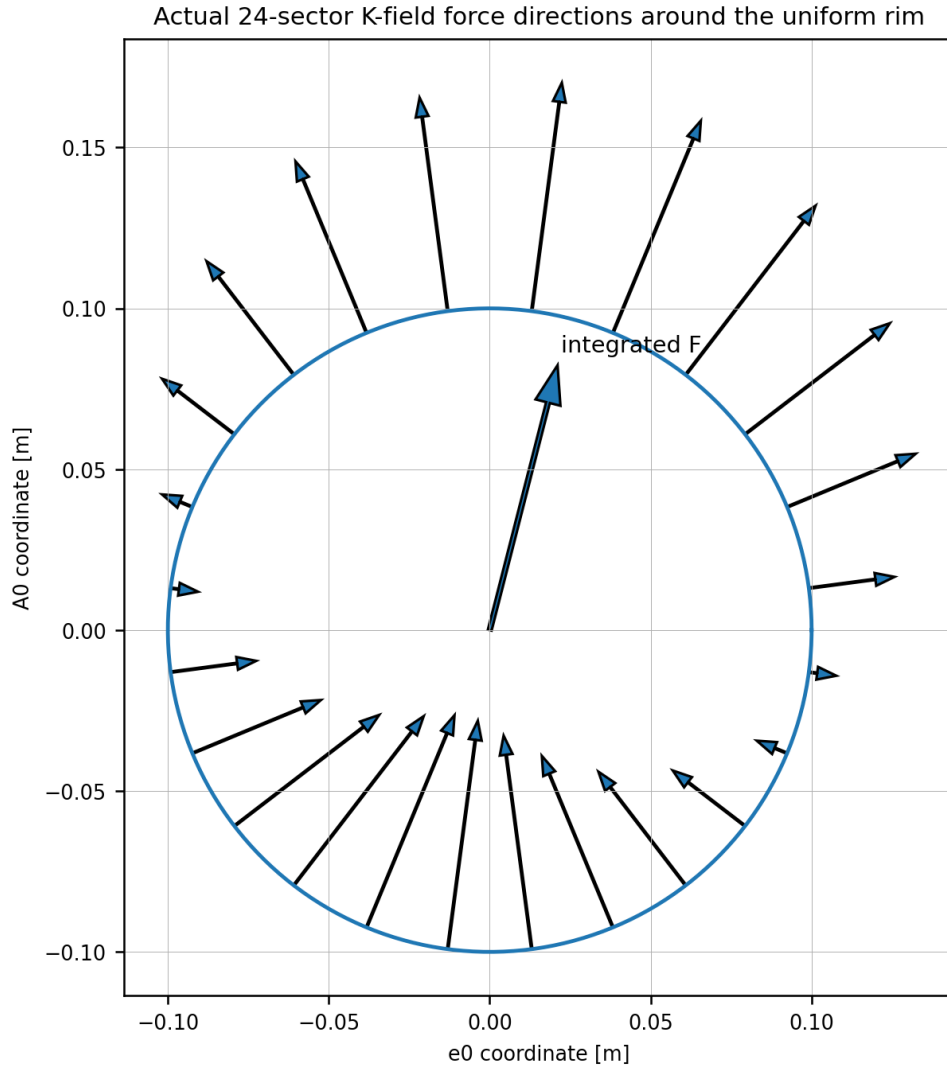


Figure 6.1. Forces computed for 24 equal rim sectors. Arrow directions are calculated from the calibrated force law; arrow lengths are uniformly enlarged for visibility. Opposite sectors contribute same-direction pair forces. The center arrow is the exact integrated resultant.

7. Exact Continuous-Rim Integration

After the rotational self term is removed, the complete rim force is the angular integral of the translation-coupled term.

$$F_{\text{ring}} = -(\kappa_F M \omega / 2\pi) \int_0^{2\pi} u(\theta) [t(\theta)^T g] d\theta \quad \text{where } g = G_{\text{phase}} V_E$$

The required second-moment integral is

$$\int_0^{2\pi} u(\theta) t(\theta)^T d\theta = \pi [e_0 A_0^T - A_0 e_0^T]$$

The bracketed antisymmetric matrix is the cross-product operator associated with the spin axis. For any unit spin axis $\mathbf{n}$, the result becomes

$$\mathbf{F}_{\text{ring}} = (\kappa_F M \omega / 2) \mathbf{n} \text{ cross } \mathbf{g}$$

This is the central closed form. Radius has disappeared because the surviving translation-coupled energy is linear in R , while the spatial derivative that converts the energy difference into force removes that factor. Radius remains in ordinary mechanical loading, energy, angular momentum, and rim speed.

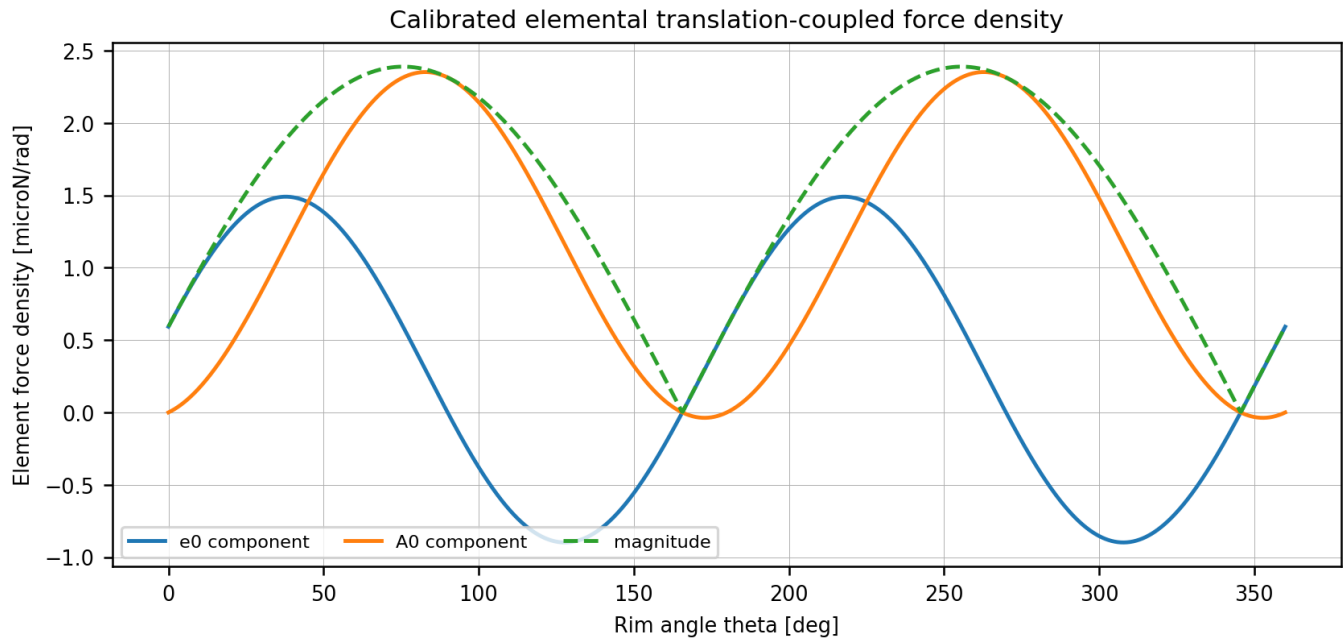


Figure 7.1. Actual calibrated force density per radian around the rim. Individual element contributions vary with angle, but the continuous angular occupancy is fixed.

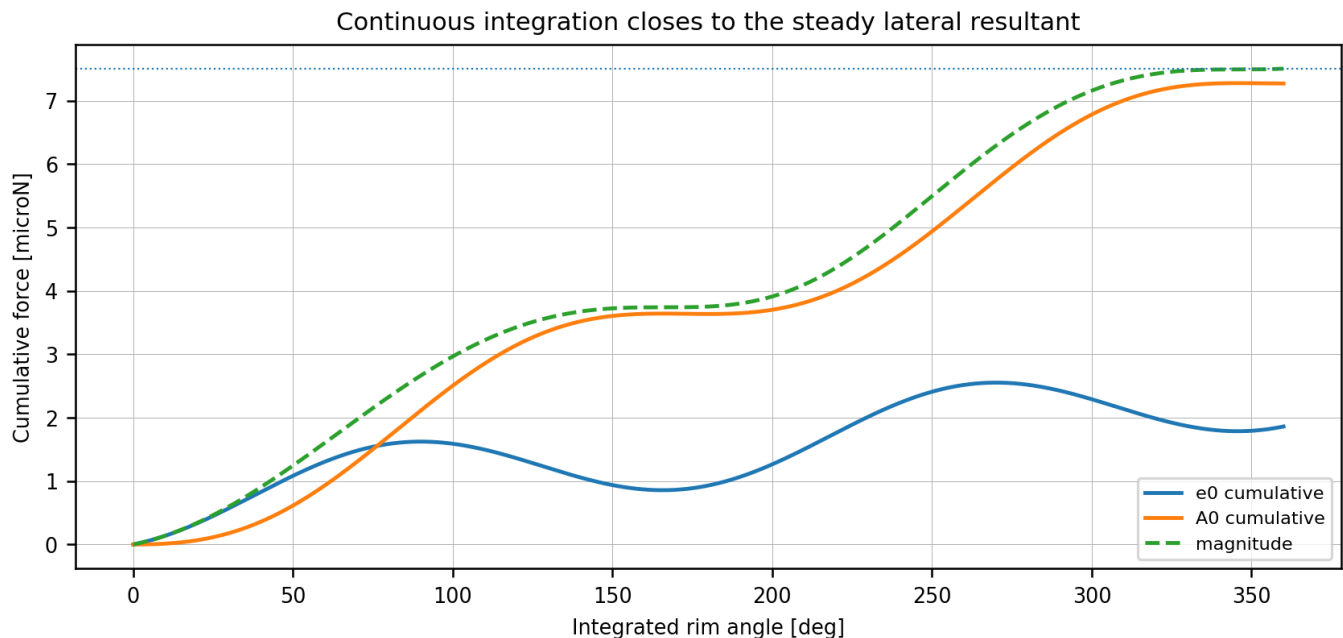


Figure 7.2. Cumulative integration from $\theta=0$ to $\theta=360$ deg. Intermediate vectors vary, while the completed circle closes to the exact steady resultant.

8. CMB-Aligned Single-Axis Result

Setting $n=c_{\text{hat}}$ gives a force perpendicular to the CMB axis by construction:

$$F_{\text{aligned}} = (\kappa_F M \omega/2) c_{\text{hat}} \text{ cross}(G_{\text{phase}} V_E)$$

The vector $c_{\text{hat}} \text{ cross } g$ is fixed in ICRS. A uniform ring simultaneously contains every angular phase, so the force does not rotate with the material. The material elements circulate, but their complete angular population remains unchanged. The integrated force is therefore steady.

Quantity	Calibrated value
Force vector [N]	[5.814375586393e-07, 5.609329067854e-06, 4.952561964817e-06]
Force magnitude	7.505365563476e-06 N = 7.505365563 microN
Calibration interval	7.428919151 to 7.581811976 microN
Force direction	RA 84.082114314 deg; Dec 41.289922334 deg
CMB-axis component	-4.789e-23 N
Period at reference rate	6.283185307180 s
Impulse per revolution	4.715760263345e-05 N s = 47.157602633 microN s

The force direction is the same lateral direction already encoded in the dual-axis average vector. For this one-kilogram uniform ring, the complete distribution converts the phase-dependent point-pair response into a constant force equal to the point pair cycle average.

9. Maximum Pure-Lateral Axis Orientation

Alignment with c_{hat} is geometrically simple, but it is not the maximum-force orientation. The general ring force magnitude is proportional to $|n \text{ cross } g|$. A force is purely lateral when its dot product with c_{hat} is zero. Both requirements are satisfied simultaneously by choosing n within the plane spanned by c_{hat} and g and making n perpendicular to g .

$$g_{\text{hat}} = g/|g| \quad n_{\text{opt}} = [c_{\text{hat}} - (c_{\text{hat}} \cdot g_{\text{hat}})g_{\text{hat}}] / |c_{\text{hat}} - (c_{\text{hat}} \cdot g_{\text{hat}})g_{\text{hat}}|$$

This axis is the component of the CMB direction that remains after removing its projection along g . Since n_{opt} is perpendicular to g , the cross product has its maximum possible magnitude $|g|$. Since n_{opt} and g lie in the $c_{\text{hat}}\text{-}g$ plane, their cross product is normal to that plane and therefore perpendicular to c_{hat} .

Quantity	Value
Optimal axis n_{opt}	[-0.315865036407, 0.646162683024, -0.694768354089]
Axis direction	RA 116.050867377 deg; Dec -44.008760899 deg
Tilt from c_{hat}	58.359138729 deg
$n_{\text{opt}} \cdot g$	4.121420e-10 m/s
Steady pure-lateral force	1.430702048308e-05 N = 14.307020483 microN
Gain over c_{hat} alignment	1.906238992 x
Force direction	RA 84.082114314 deg; Dec 41.289922334 deg

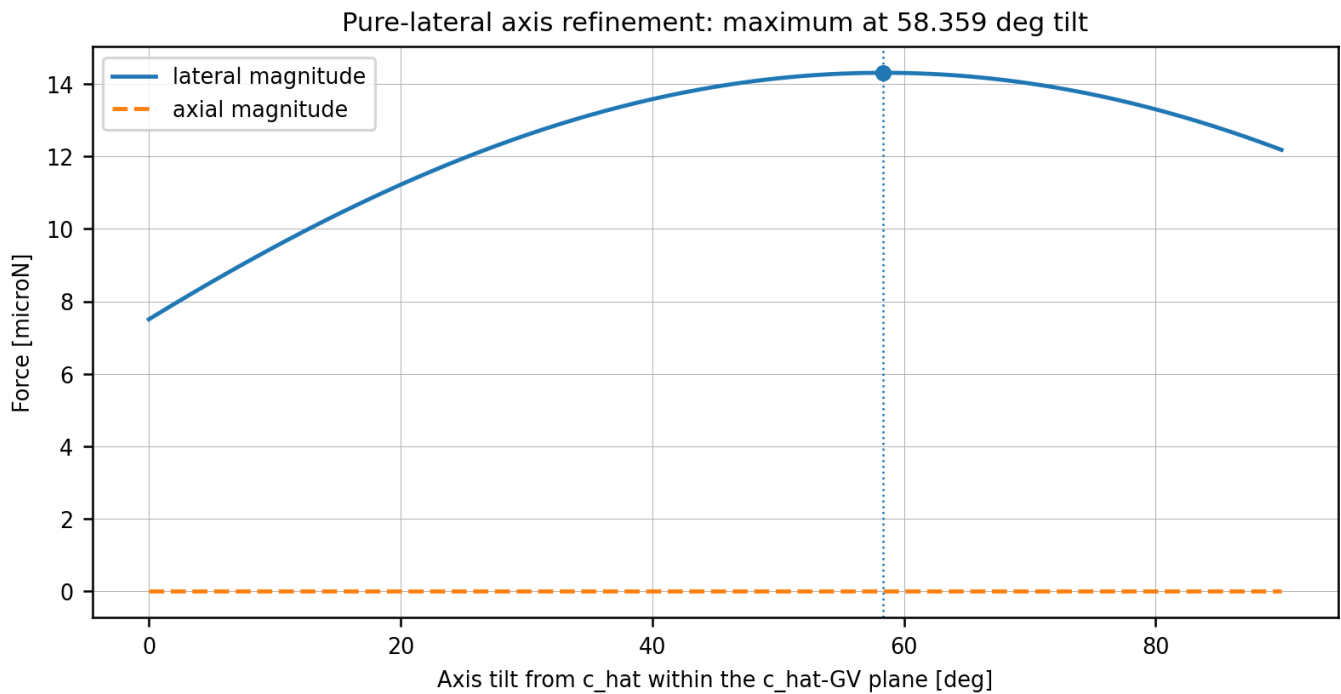


Figure 9.1. Force as the axis is tilted from $\hat{c}$ within the plane containing $\hat{c}$ and $\mathbf{g}$. The force remains lateral throughout this plane; its magnitude reaches the global pure-lateral maximum when the axis is perpendicular to $\mathbf{g}$.

10. Comparison with Point-Mass and Dual-Axis Devices

A uniform ring and a balanced antipodal point pair use the same underlying sign relation. Their difference is angular occupancy. A point pair occupies only two opposite phases at a time, so its force pulses during rotation. A uniform ring occupies all phases simultaneously, so the oscillating parts cancel and the cycle-average vector remains continuously present.

Device	Active mass	Force at base $\omega=1$ rad/s	Cycle average	Character
Uniform ring, axis = $\hat{c}$	1.000 kg	7.505365563 microN	7.505365563 microN	steady
Uniform ring, maximum pure-lateral axis	1.000 kg	14.307020483 microN	14.307020483 microN	steady
Single-axis antipodal point pair, total mass 1 kg, axis = $\hat{c}$	1.000 kg	15.010731127 microN	7.505365563 microN	pulsating
Dual-axis 2:1 reference, two 1 kg masses	2.000 kg	127.068061511 microN	7.505365563 microN	strongly modulated

The CMB-aligned one-kilogram ring produces the same 7.505 microN average lateral force as the two-kilogram dual-axis reference at the same base angular rate. The dual-axis motion produces a much larger instantaneous peak, but its compound trajectory also introduces strong cancellation over the cycle. The optimized single-axis ring produces 14.307 microN continuously at the same one-kilogram mass and angular rate.

This comparison separates peak force from useful continuous force. The dual-axis device is a peak-generating geometry. The single-axis continuous rim is a steady-force geometry. For minimum structural complexity and maximum average force per unit active mass, the continuous single-axis rim is the more efficient arrangement under the present equations.

11. Radius, Mass, and Spin Scaling

At fixed angular rate, the exact K-field resultant is independent of radius and linear in active mass and angular rate:

F_{ring} proportional to $\kappa_F M \omega$ F_{ring} independent of R when ω is fixed

The CMB-aligned reference scaling is

$$F_{\text{aligned}} = 7.505365563 \text{ microN (M/1 kg)}(\omega/1 \text{ rad/s}) = 0.785960044 \text{ microN (M/1 kg)(RPM/1 rpm)}$$

The maximum pure-lateral scaling is

$$F_{\text{max_lat}} = 14.307020483 \text{ microN (M/1 kg)}(\omega/1 \text{ rad/s}) = 1.498240365 \text{ microN (M/1 kg)(RPM/1 rpm)}$$

At fixed rim speed $v=R \omega$, the force varies as v/R . A smaller ring at the same rim speed rotates faster and therefore produces more force. At fixed angular momentum or fixed stored energy, the radius dependence is different because ω itself changes with R .

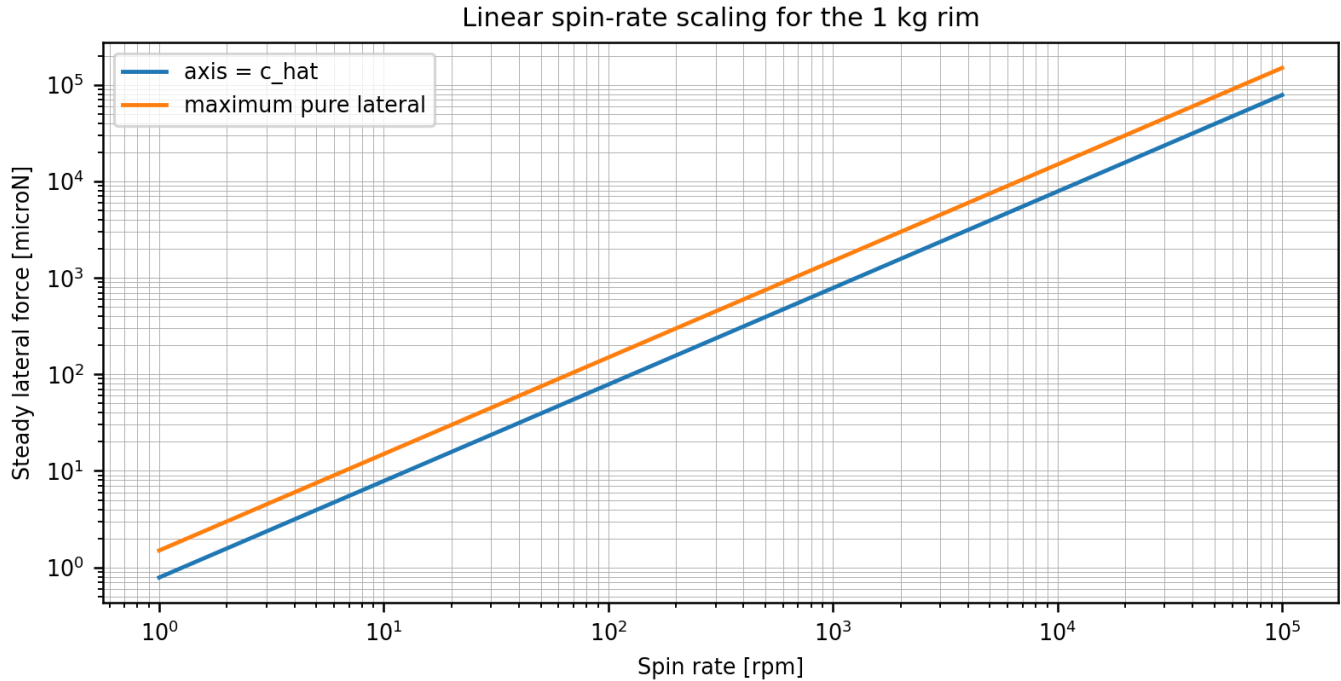


Figure 11.1. Exact linear force scaling over five decades of spin rate. Both curves use the requested one-kilogram mass.

RPM	ω [rad/s]	Rim speed [m/s]	Aligned force	Max pure-lateral force
1	0.10472	0.010472	0.78596 microN	1.49823 microN
9.5493	1	0.1	7.50537 microN	14.307 microN
10	1.0472	0.10472	7.8596 microN	14.9823 microN
100	10.472	1.0472	78.596 microN	149.823 microN
1000	104.72	10.472	785.96 microN	1498.23 microN
10000	1047.2	104.72	7859.6 microN	14982.3 microN
100000	10472	1047.2	78596 microN	149823 microN

12. Mechanical Loading and Material Economy

The K-field force is independent of radius at fixed ω , but ordinary mechanical demands are not. For a thin ring, moment of inertia and stored energy scale as R^2 , while ideal hoop tension scales as R . Both tension and energy rise as ω^2 . The K-field force rises only as ω . Consequently, increasing RPM eventually produces a rapidly worsening ratio of mechanical load to useful force.

$$I = M R^2 \quad E_{\text{rot}} = 0.5 M R^2 \omega^2 \quad L = M R^2 \omega \quad T_{\text{hoop}} = M R \omega^2 / (2\pi)$$

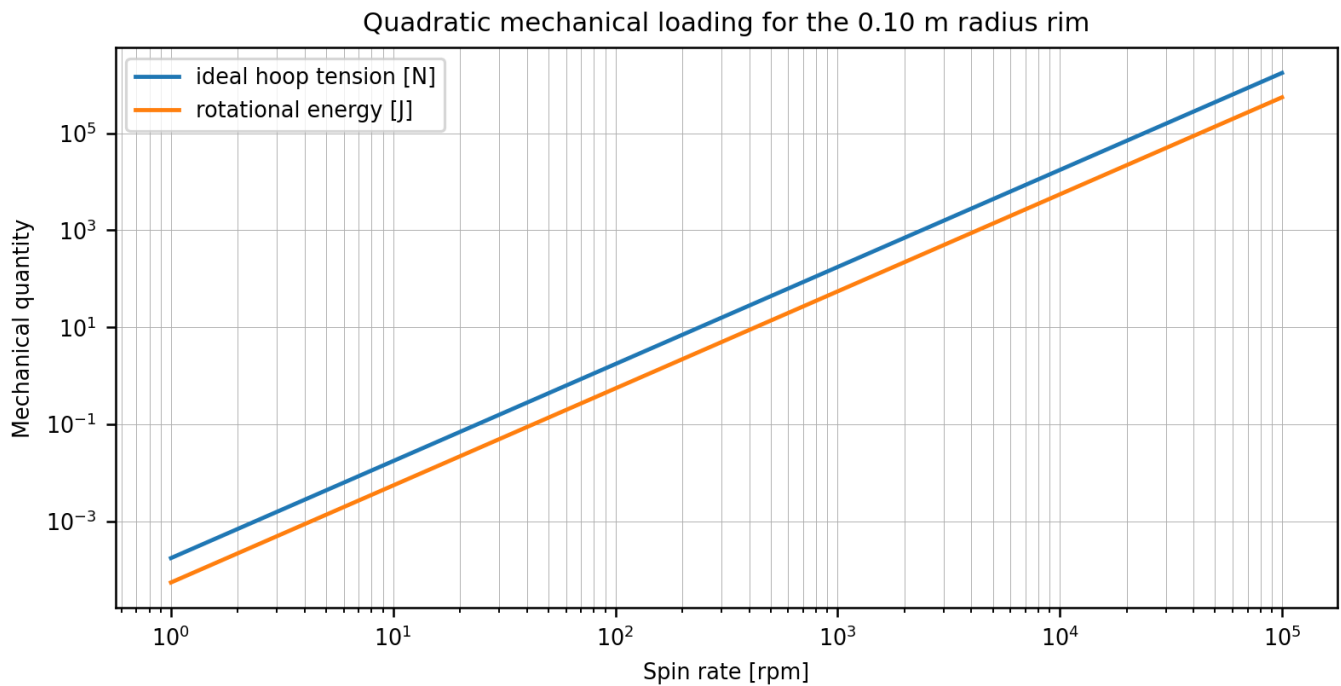


Figure 12.1. Mechanical loading for the requested 0.10 m radius rim. Hoop tension and rotational energy rise quadratically with RPM.

RPM	Rim speed [m/s]	Hoop tension [N]	Rotational energy [J]	Angular momentum [kg m ² /s]
1	0.010472	0.000174533	5.48311e-05	0.0010472
9.5493	0.1	0.0159155	0.005	0.01
10	0.10472	0.0174533	0.00548311	0.010472
100	1.0472	1.74533	0.548311	0.10472
1000	10.472	174.533	54.8311	1.0472
10000	104.72	17453.3	5483.11	10.472
100000	1047.2	1.74533e+06	548311	104.72

For a minimum-material experimental device, radius should therefore be set by bearings, motor clearance, sensing geometry, and containment rather than by a desire to increase K-field force. Dense rim material can reduce physical cross-section, but it does not increase force per kilogram. The primary efficiency measure is active rim mass divided by total device mass.

13. Direction, Spin Reversal, and Laboratory Orientation

The force direction is fixed by the directed spin axis and the anisotropic kernel g . Reversing spin changes n to $-n$ and reverses the force without changing its magnitude. This provides a direct signed experimental control.

$$F(-n) = -F(n)$$

For the CMB-aligned and optimized pure-lateral axes, the force points along the same ICRS direction: RA 84.0821143 deg, Dec 41.2899223 deg for the selected positive spin branch. The opposite spin branch points to the antipodal direction. A terrestrial apparatus that is fixed to the laboratory floor does not remain fixed in ICRS; maintaining a constant inertial force direction requires tracking the locked axis, while a non-tracking apparatus produces a predictable orientation modulation in laboratory coordinates.

14. Numerical Reconstruction and Validation

The analytic integral was independently reconstructed by dividing the rim into equal midpoint sectors. The complete element law was evaluated for each sector, including the radius-dependent self term, and all force and torque vectors were summed. The reconstruction converges to the exact closed form; the self term and torque remain at numerical roundoff.

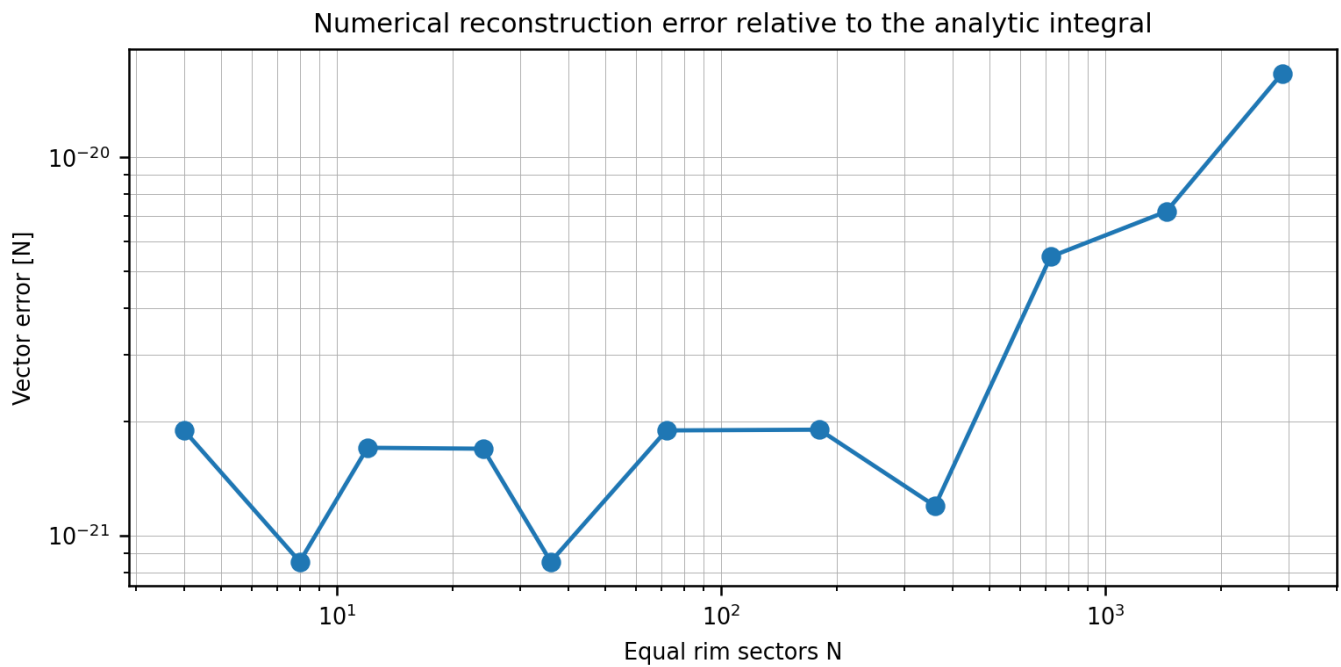


Figure 14.1. Vector error of equal-sector midpoint reconstruction relative to the analytic continuous-rim result.

Segments	Force error [N]	Self-term residual [N]	Torque residual [N m]
4	1.897e-21	2.285e-27	1.413e-23
8	8.536e-22	1.212e-27	8.205e-24
12	1.707e-21	8.598e-28	4.810e-24
24	1.697e-21	2.427e-27	7.991e-24
36	8.536e-22	1.365e-27	7.249e-24
72	1.897e-21	7.274e-28	2.007e-24
180	1.906e-21	1.273e-27	4.793e-24
360	1.198e-21	5.521e-27	1.366e-24
720	5.461e-21	5.369e-27	1.446e-24
1440	7.187e-21	2.708e-27	7.797e-25
2880	1.664e-20	9.379e-27	9.140e-25

The validation also confirms symmetry of G_{phase} , orthonormality and right-handedness of the $e_0\text{-}A_0\text{-}c_{\text{hat}}$ frame, perpendicularity of the reported lateral forces to c_{hat} , and the exact maximum-force condition $n_{\text{opt}} \cdot g = 0$.

15. Physical Interpretation in Common Language

The flywheel works as a complete set of paired moving masses. Every small piece of rim has a partner directly across the center. The two pieces move in opposite tangential directions. Because the K-field response depends on direction through $G_{\text{phase}} V_E$, their response signs are opposite. Their locations are also opposite. Those two sign reversals make the two physical force arrows point together.

A two-mass dumbbell contains only one antipodal pair, so its force grows, shrinks, and changes its internal distribution as the pair rotates. A continuous rim contains every possible antipodal pair angle at the same time. The changing parts balance one another, while the common directional part survives. The result is a steady force.

The 10 cm radius determines how fast the material moves, how much rotational energy is stored, and how much tension the rim must carry. It does not determine the K-field force when angular speed is held fixed. The active quantities are total rim mass, angular speed, spin-axis direction, the Earth-CMB translation state, the phase metric, and the calibrated coupling

constant.

16. Principal Findings

1. A dual-axis mechanism is not required. A single fixed-axis uniform rim produces a nonzero steady lateral force under the same translation-coupled K-field law.
2. Uniform angular mass occupancy does not cancel the lateral force. Opposite rim elements add because both tangential response and radial position reverse.
3. The radius-dependent rotational self term cancels exactly over antipodal elements. The surviving net force is independent of radius at fixed angular rate.
4. The exact continuous-rim law is $F_{\text{ring}} = (\kappa_F M \omega / 2) \mathbf{n} \times (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$.
5. For the requested 1 kg, 0.10 m radius ring at $\omega = 1$ rad/s with $\mathbf{n} = \mathbf{c}_{\text{hat}}$, the force is steady, purely lateral, and equal to 7.5053656 microN.
6. The maximum steady force that remains purely lateral is 14.3070205 microN at the same mass and angular rate. Its axis is tilted 58.3591387 deg from $\mathbf{c}_{\text{hat}}$ and is perpendicular to $\mathbf{G}_{\text{phase}} \mathbf{V}_E$.
7. The optimized single-axis rim produces more cycle-average lateral force per kilogram than the calibrated dual-axis reference, while eliminating the gimbal and the second rotation drive.
8. Force rises linearly with RPM, whereas hoop tension and stored energy rise quadratically. High RPM is therefore mechanically expensive relative to mass scaling or parallel synchronized rotors.

17. Conclusion

The continuous-rim calculation reduces the lateral-force mechanism to its simplest complete geometry. A one-kilogram uniform ring rotating about one fixed axis is sufficient. Antipodal symmetry removes the radius-dependent self term but preserves the translation-coupled force because the direction and response sign reverse together. Continuous angular occupancy converts a pulsating point-pair force into a steady vector.

For the specified 0.10 m radius and reference rate of 1 rad/s, CMB-axis alignment produces 7.5053656 microN. Reorienting the same rotor to the exact maximum pure-lateral axis increases the force to 14.3070205 microN without changing mass, radius, or speed. The resulting architecture is materially compact: one uniform rim, one shaft, one bearing set, one drive, and a fixed inertial orientation. The machine-readable companion preserves every input, equation, Cell3 node, plane, sector force, scaling state, and validation result required for reconstruction.

Appendix A. Complete 27-Node Cell3 Ledger

Each node is defined by half-step primitive coefficients ($i/2, j/2, k/2$), integer index (i, j, k), ICRS native coordinate, and sheet grade $\lambda = i + j + k$.

i	j	k	x [native]	y [native]	z [native]	lambda
-1	-1	-1	-0.010974902500	0.007385983127	0.039661911012	-3
-1	-1	0	-0.011290092243	-0.004742877881	0.018199733726	-2
-1	-1	1	-0.011605281985	-0.016871738890	-0.003262443559	-1
-1	0	-1	-0.009875529516	0.013837068257	0.022628660244	-2
-1	0	0	-0.010190719258	0.001708207248	0.001166482958	-1
-1	0	1	-0.010505909001	-0.010420653760	-0.020295694327	0
-1	1	-1	-0.008776156532	0.020288153386	0.005595409476	-1
-1	1	0	-0.009091346274	0.008159292378	-0.015866767810	0
-1	1	1	-0.009406536017	-0.003969568631	-0.037328945095	1
0	-1	-1	-0.000784183241	0.005677775879	0.038495428054	-2
0	-1	0	-0.001099372984	-0.006451085129	0.017033250768	-1
0	-1	1	-0.001414562727	-0.018579946138	-0.004428926518	0
0	0	-1	0.000315189743	0.012128861009	0.021462177286	-1
0	0	0	0.000000000000	0.000000000000	0.000000000000	0
0	0	1	-0.000315189743	-0.012128861009	-0.021462177286	1
0	1	-1	0.001414562727	0.018579946138	0.004428926518	0
0	1	0	0.001099372984	0.006451085129	-0.017033250768	1
0	1	1	0.000784183241	-0.005677775879	-0.038495428054	2
1	-1	-1	0.009406536017	0.003969568631	0.037328945095	-1
1	-1	0	0.009091346274	-0.008159292378	0.015866767810	0
1	-1	1	0.008776156532	-0.020288153386	-0.005595409476	1
1	0	-1	0.010505909001	0.010420653760	0.020295694327	0
1	0	0	0.010190719258	-0.001708207248	-0.001166482958	1
1	0	1	0.009875529516	-0.013837068257	-0.022628660244	2
1	1	-1	0.011605281985	0.016871738890	0.003262443559	1
1	1	0	0.011290092243	0.004742877881	-0.018199733726	2
1	1	1	0.010974902500	-0.007385983127	-0.039661911012	3

Appendix B. Cell3 Vertices and Boundary Planes

The eight vertices are the coefficient combinations (+/-0.5, +/-0.5, +/-0.5).

a coeff	b coeff	c coeff	x [native]	y [native]	z [native]
-0.5	-0.5	-0.5	-0.010974902500	0.007385983127	0.039661911012
-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443559
-0.5	0.5	-0.5	-0.008776156532	0.020288153386	0.005595409476
-0.5	0.5	0.5	-0.009406536017	-0.003969568631	-0.037328945095
0.5	-0.5	-0.5	0.009406536017	0.003969568631	0.037328945095
0.5	-0.5	0.5	0.008776156532	-0.020288153386	-0.005595409476
0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443559
0.5	0.5	0.5	0.010974902500	-0.007385983127	-0.039661911012

The six boundary planes use $n \cdot x + d = 0$.

Plane family	Side	Unit normal [ICRS]	d [native]
ab faces (constant c coefficient)	positive	[0.194092201, 0.913177169, 0.358379235]	-0.018828573500193
ac faces (constant b coefficient)	positive	[0.089052594, 0.866576769, -0.491033948]	0.014052166903391
bc faces (constant a coefficient)	positive	[-0.995965190, 0.083602163, -0.032619291]	-0.010254361619054
ab faces (constant c coefficient)	negative	[0.194092201, 0.913177169, 0.358379235]	0.018828573500193
ac faces (constant b coefficient)	negative	[0.089052594, 0.866576769, -0.491033948]	-0.014052166903391
bc faces (constant a coefficient)	negative	[-0.995965190, 0.083602163, -0.032619291]	0.010254361619054

Appendix C. Formula and Identity Ledger

Name	Expression
Rim radial unit	$u(\theta) = \cos(\theta)e_0 + \sin(\theta)A_0$
Rim tangent	$t(\theta) = -\sin(\theta)e_0 + \cos(\theta)A_0$
Position	$r(\theta) = R u(\theta)$
Mass measure	$dm = (M/2\pi)d\theta$
Velocity	$v(\theta) = V_E + R \omega t(\theta)$
Translation kernel	$B(\theta) = t(\theta)^T G_{\text{phase}} V_E$
Self kernel	$C(\theta) = t(\theta)^T G_{\text{phase}} t(\theta)$
Element force	$dF = -\kappa_F dm [\omega B + R \omega^2 C] u$
Antipodal sum	$dF(\theta) + dF(\theta + \pi) = -2 \kappa_F dm \omega B u$
Continuous rim	$F = (\kappa_F M \omega / 2) n \text{ cross}(G_{\text{phase}} V_E)$
CMB-aligned	$F = (\kappa_F M \omega / 2) \hat{c} \text{ cross}(G_{\text{phase}} V_E)$
Optimal axis	$n_{\text{opt}} = [\hat{c} - (\hat{c} \cdot \hat{g}) \hat{g}] / \text{norm}[\dots]$
Maximum pure lateral	$F_{\text{max}} = (\kappa_F M \omega / 2) G_{\text{phase}} V_E $
Torque	$d_{\tau} = r \text{ cross } dF = 0$
Spin reversal	$F(-n) = -F(n)$

Appendix D. Thirty-Degree Element Ledger

The table samples the machine-readable one-degree midpoint ledger at 30-degree intervals. Forces are for one 1/360-rim sector and use the calibrated coupling.

theta [deg]	B [m/s]	C	dF translation [pN]	dF self [pN]
0.5	-125179.081329	1.646669058	10689.079655	0.014061
30.5	-344401.672118	3.443547601	29408.563056	0.029405
60.5	-471342.112990	6.896547632	40248.045736	0.058890
90.5	-471986.815329	8.552669121	40303.097064	0.073032
120.5	-346163.031662	6.755790578	29558.966081	0.057688
150.5	-127585.143211	3.302790547	10894.534008	0.028203
180.5	125179.081329	1.646669058	10689.079655	0.014061
210.5	344401.672118	3.443547601	29408.563056	0.029405
240.5	471342.112990	6.896547632	40248.045736	0.058890
270.5	471986.815329	8.552669121	40303.097064	0.073032
300.5	346163.031662	6.755790578	29558.966081	0.057688
330.5	127585.143211	3.302790547	10894.534008	0.028203

Appendix E. Source Lock and Reconstruction Metadata

Item	Value
Dual-axis source file	K-Field_dual_axis_kinetic_mass_CMB_traversal_lateral_force_comptation_1784803951(2).json
Dual-axis SHA-256	70d65b1fc297949d92be2b18d3a146ac4f1e008da155ee3dd7df731b1d8fb310
Earth coupling source file	K-Field_earth_kinematic_derivation_of_coupling_constant_1784828045(2).json
Earth coupling SHA-256	2f9d605451b26f29a84ff7511dd69efa3d53db6212a586b0b6562dc54faf1477
Report epoch	1784896442
Generated UTC	2026-07-24T12:34:02Z
Machine-readable companion	K-Field_laerl_force_analysis_1784896442.json
Coordinate frame	ICRS/J2000 Cartesian
Cell3 origin	(0,0,0)
Numerical integration	Equal-sector midpoint quadrature; analytic closed form authoritative